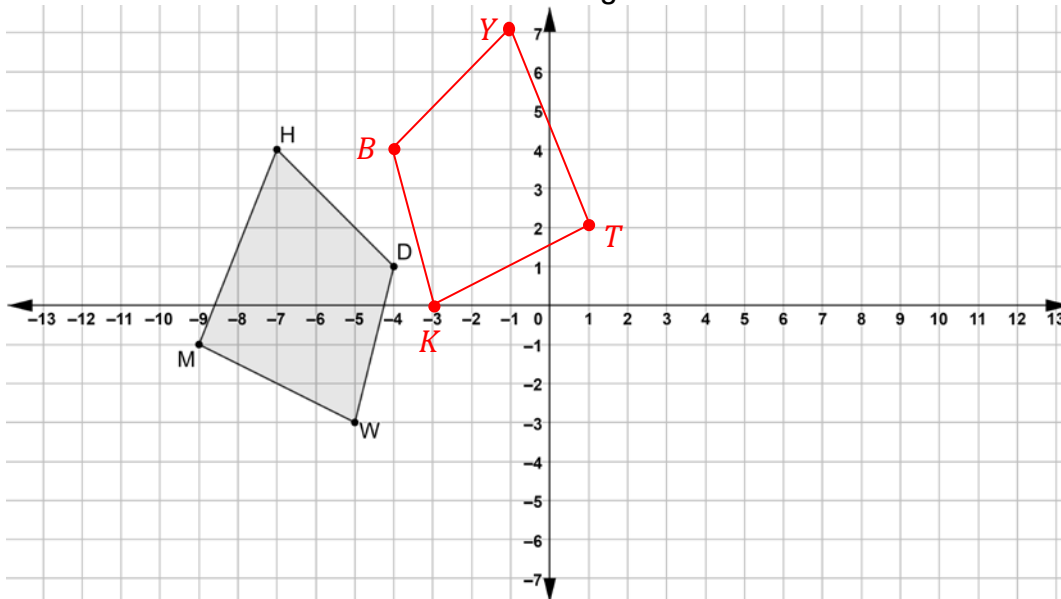


Quadrilateral $HDWM$ is shown on the coordinate grid.



1. Draw the quadrilateral that is the result of a translation right 8 units and up 3 units and then reflected across the y -axis. Label the new quadrilateral $YBKT$.

Complete the table.

	Transformation	Mapping	Algebraic Description
2.	translation	$HDWM \rightarrow H'D'W'M'$	$(x, y) \rightarrow (x + 8, y + 3)$
3.	reflection	$H'D'W'M' \rightarrow YBKT$	$(x, y) \rightarrow (-x, y)$

4. Complete the table of values for each quadrilateral.

Quadrilateral $HDWM$			Quadrilateral $H'D'W'M'$			Quadrilateral $YBKT$		
Vertex	x	y	Vertex	x	y	Vertex	x	y
H	-7	4	H'	1	7	Y	-1	7
D	-4	1	D'	4	4	B	-4	4
W	-5	-3	W'	3	0	K	-3	0
M	-9	-1	M'	-1	2	T	1	2

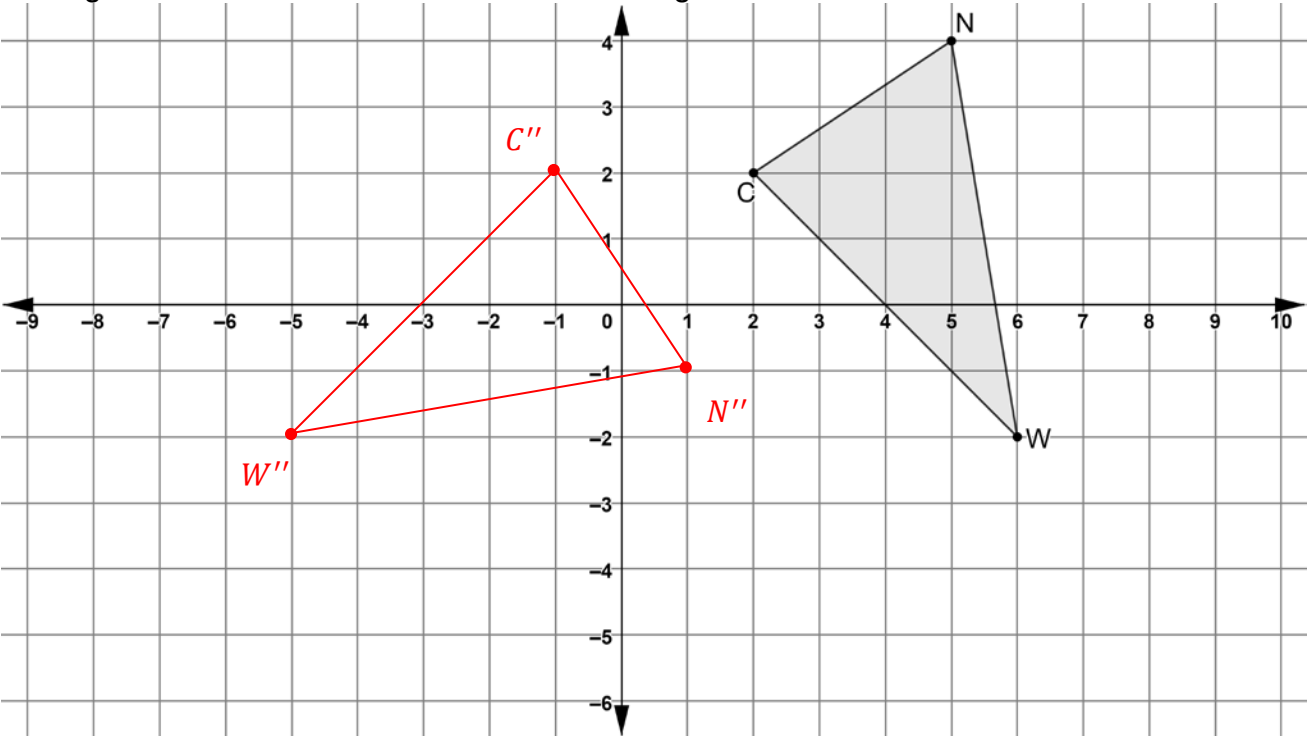
5. Explain how the tables for quadrilaterals $HDWM$ and $H'D'W'M'$ show the effect of the transformation on the coordinates that result in the algebraic description.

Each x -coordinate increases by 8, each y -coordinate increases by 3.

6. Explain how the tables for quadrilaterals $H'D'W'M'$ and $YBKT$ show the effect of the transformation on the coordinates that result in the algebraic description.

The x -coordinate changes its sign.

Triangle NWC is shown on the coordinate grid.



7. Draw the triangle that is the result of a rotation 90° clockwise about the origin and then a translation of left 3 units and up 4 units. Label the new triangle $N''W''C''$.

Complete the table.

	Transformation	Mapping	Algebraic Description
8.	rotation 90° clockwise	$\triangle NWC \rightarrow \triangle N'W'C'$	$(x, y) \rightarrow (y, -x)$
9.	translation	$\triangle N'W'C' \rightarrow \triangle N''W''C''$	$(x, y) \rightarrow (x - 3, y + 4)$

10. Describe the connection between the algebraic description of the rotation and the vertices of $\triangle NWC$ and $\triangle N'W'C'$.

The coordinates change spots and the second (now y -coordinate) changes sign.